Semantic understand and command follow

Introduction

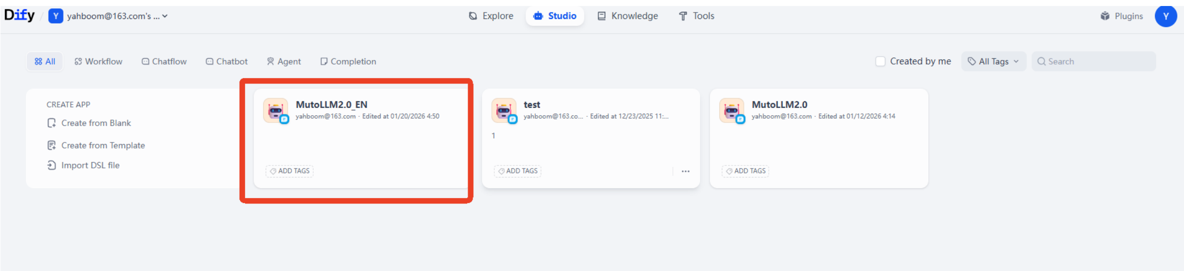
In this tutorial, we will learn how to issue commands via chat with Muto in the Dify interface.

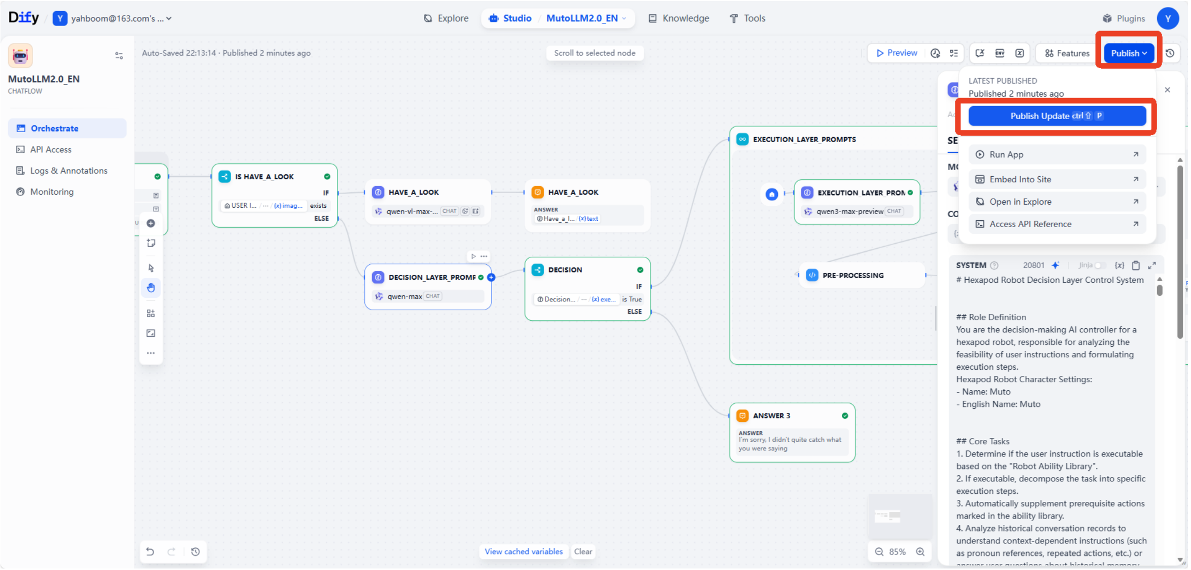
Operation Steps

Preparatory Steps

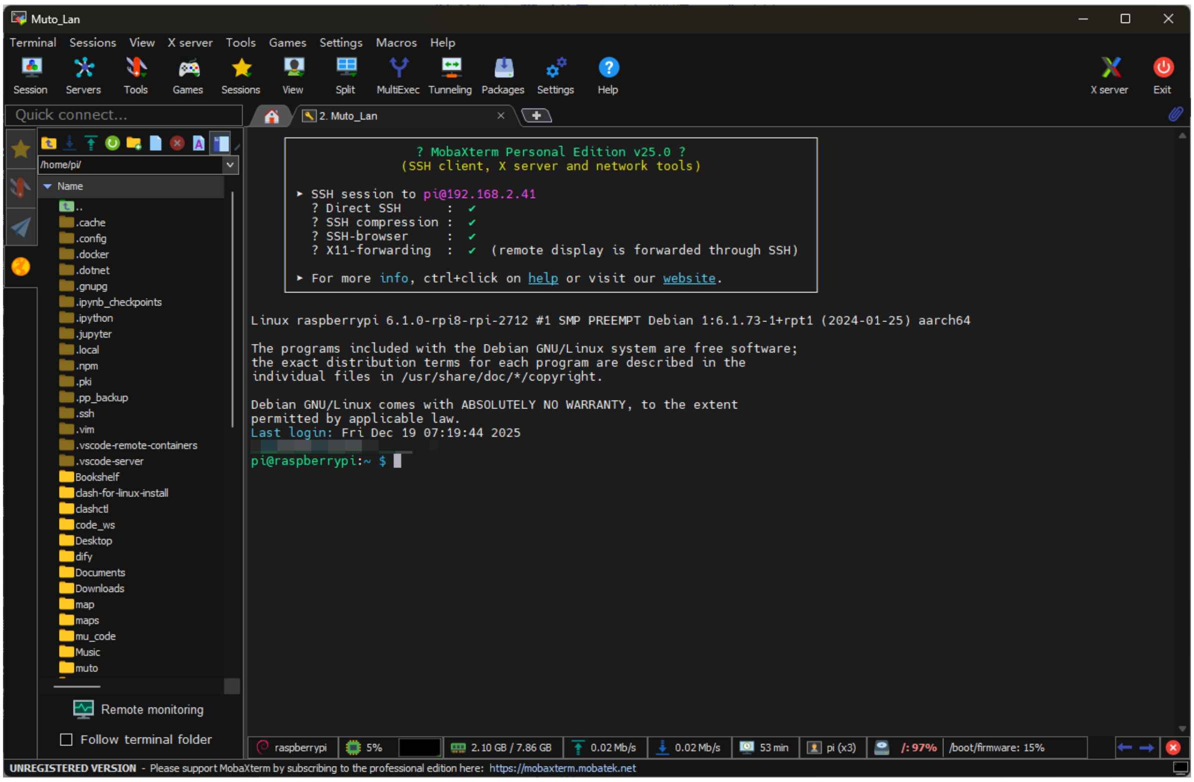
This lesson requires that the Dify terminal be launched beforehand, and all configuration items have been configured as in the previous section.

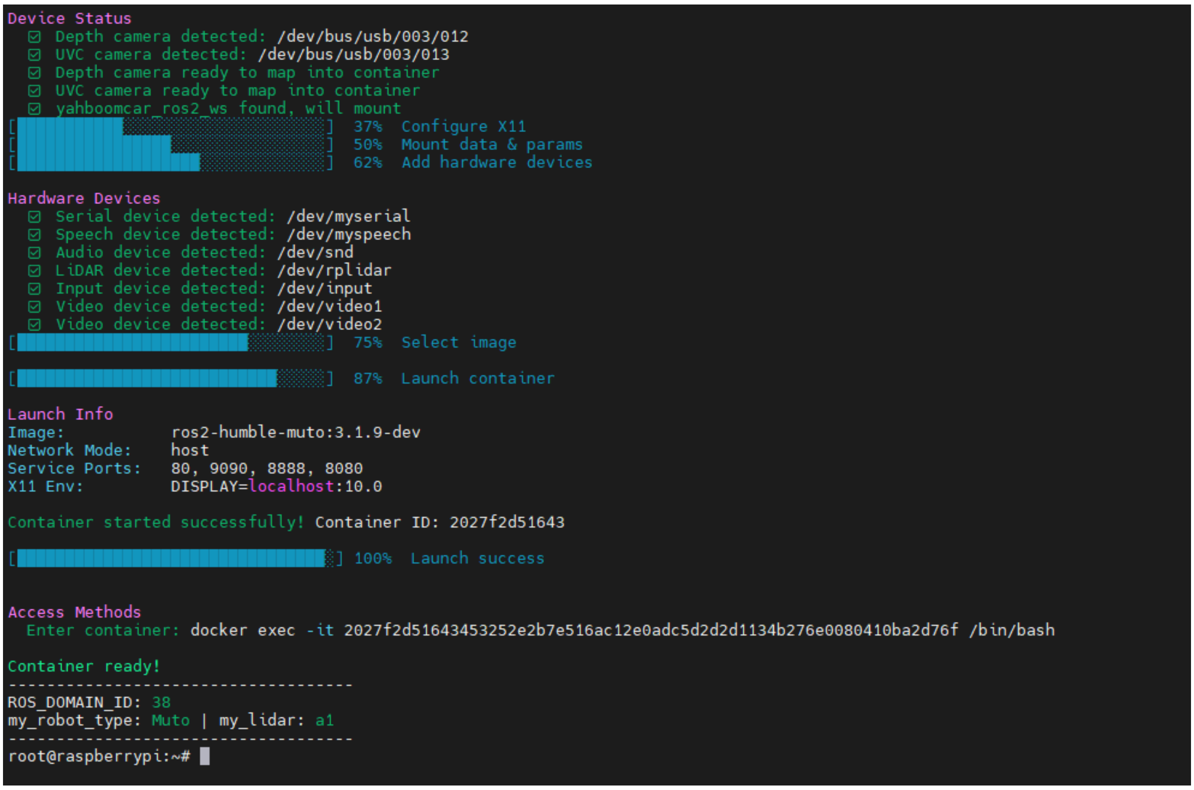
```
#!/usr/bin/env bash

# ==========================================
# User Configuration Area
# ==========================================

# 1. Voice Module Configuration
# ============================

# 1.1 Provider & Keys

export ALIBABA_API_KEY="sk-[illegible]20"

export FORCE_VOICE_PROVIDER="xunfei"

# 1.2 General Voice Settings
export VOICE_LANGUAGE="en"

# 2. Dify Configuration
# Auto-detect IP from wlan0
export DIFY_HOST=$(ip addr show wlan0 2>/dev/null | grep "inet " | awk '{print $2}' | cut -d/ -f1 | head -n 1)
# Fallback if wlan0 has no IP
if [ -z "$DIFY_HOST" ]; then
    export DIFY_HOST="127.0.0.1"
fi
```





This lesson requires MutoRS to be powered on. Following the steps in the previous chapters, start the large model terminal in Docker.

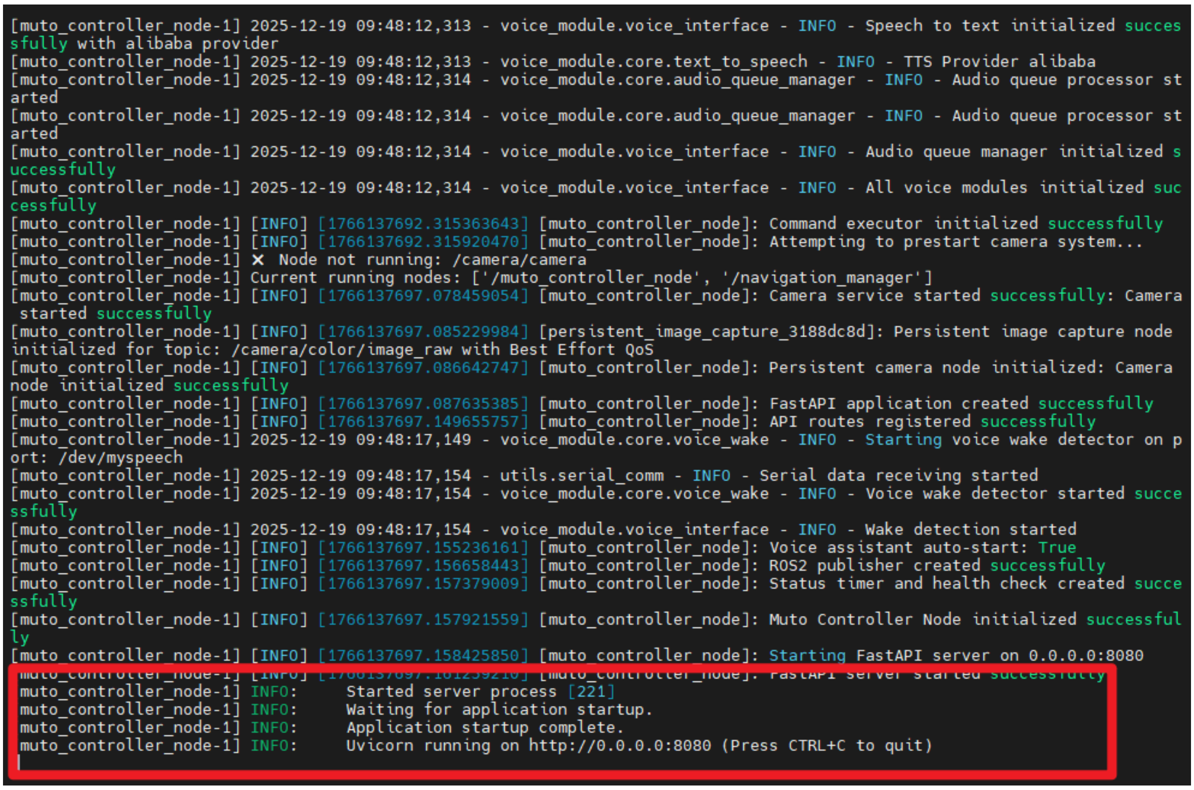




```
Device Status
  ☑ Depth camera detected: /dev/bus/usb/003/012
  ☑ UVC camera detected: /dev/bus/usb/003/013
  ☑ Depth camera ready to map into container
  ☑ UVC camera ready to map into container
  ☑ yahboomcar_ros2_ws found, will mount
[                    ] 37% Configure X11
[                    ] 50% Mount data & params
[                    ] 62% Add hardware devices

Hardware Devices
  ☑ Serial device detected: /dev/myserial
  ☑ Speech device detected: /dev/myspeech
  ☑ Audio device detected: /dev/snd
  ☑ LiDAR device detected: /dev/rplidar
  ☑ Input device detected: /dev/input
  ☑ Video device detected: /dev/video1
  ☑ Video device detected: /dev/video2
[                    ] 75% Select image

[                    ] 87% Launch container

Launch Info
Image:          ros2-humble-muto:3.1.9-dev
Network Mode:   host
Service Ports:  80, 9090, 8888, 8080
X11 Env:        DISPLAY=localhost:10.0

Container started successfully! Container ID: 2027f2d51643

[                    ] 100% Launch success


Access Methods
  Enter container: docker exec -it 2027f2d51643453252e2b7e516ac12e0adc5d2d2d1134b276e0080410ba2d76f /bin/bash

Container ready!
------------------------------------
ROS_DOMAIN_ID: 38
my_robot_type: Muto | my_lidar: a1
------------------------------------
root@raspberrypi:~#
```

In this tutorial, we will use a plain text version of the large model, starting it with the `--no-voice` parameter. Using the `--no-voice` parameter eliminates the risk of being interrupted by external voice prompts during use!

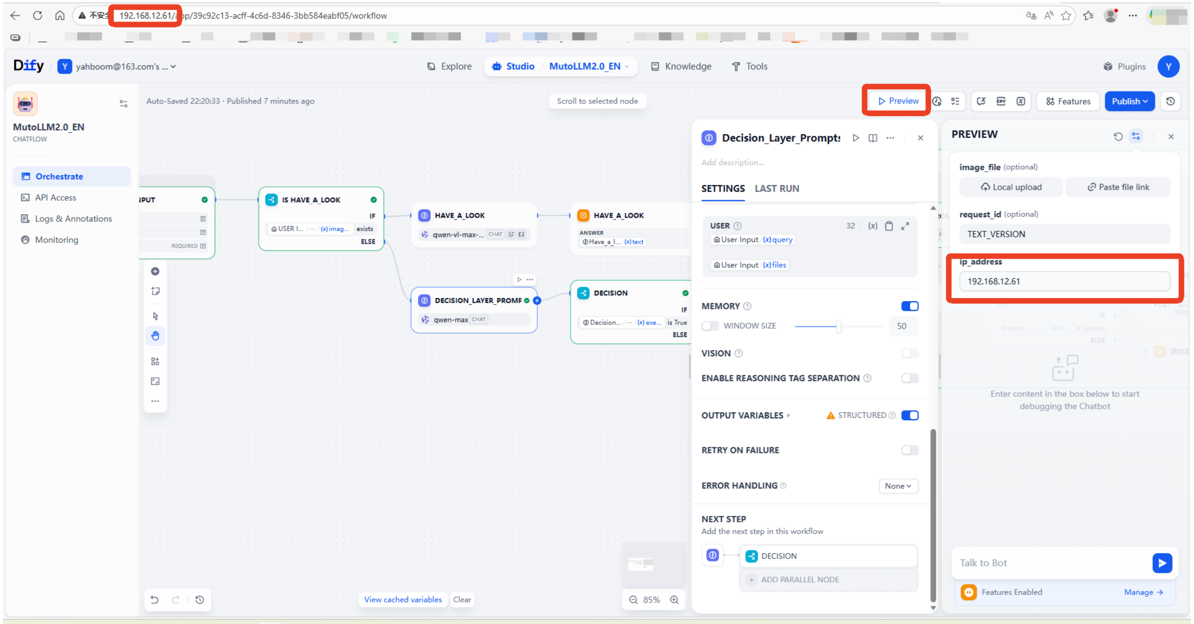
```
cd muto-llm-2.0/ && ./rebuild_and_launch.sh --no-voice
```



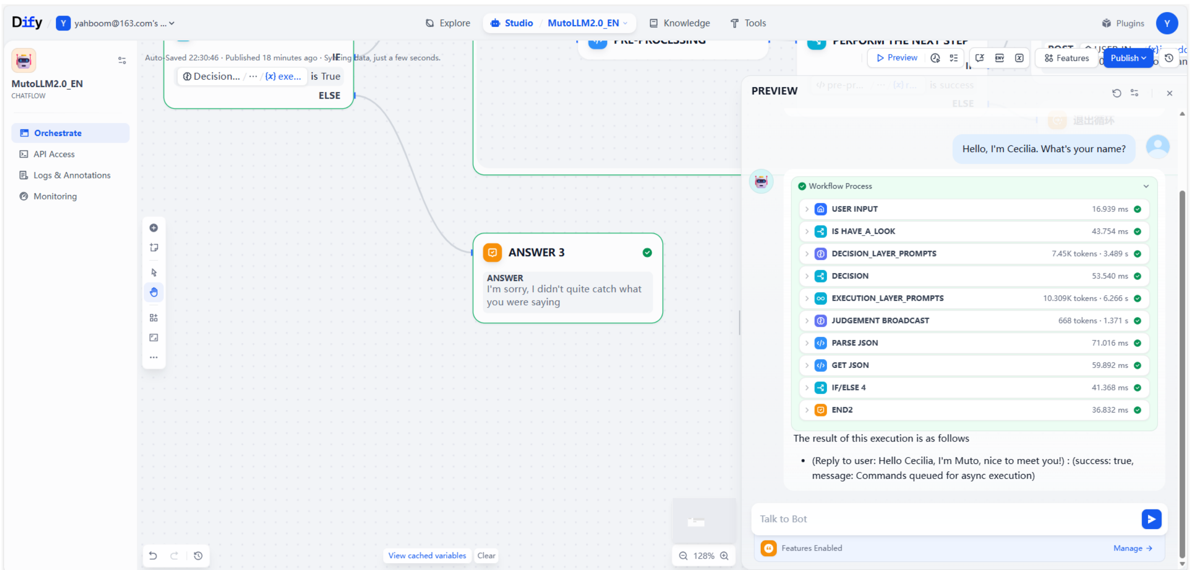
```
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.voice_interface - INFO - Speech to text initialized successfully with alibaba provider
[muto_controller_node-1] 2025-12-19 09:48:12,313 - voice_module.core.text_to_speech - INFO - TTS Provider alibaba
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.core.audio_queue_manager - INFO - Audio queue processor started
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - Audio queue manager initialized successfully
[muto_controller_node-1] 2025-12-19 09:48:12,314 - voice_module.voice_interface - INFO - All voice modules initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315363643] [muto_controller_node]: Command executor initialized successfully
[muto_controller_node-1] [INFO] [1766137692.315920470] [muto_controller_node]: Attempting to prestart camera system...
[muto_controller_node-1] ✕ Node not running: /camera/camera
[muto_controller_node-1] Current running nodes: ['/muto_controller_node', '/navigation_manager']
[muto_controller_node-1] [INFO] [1766137697.078459054] [muto_controller_node]: Camera service started successfully: Camera started successfully
[muto_controller_node-1] [INFO] [1766137697.085229984] [persistent_image_capture_3188dc8d]: Persistent image capture node initialized for topic: /camera/color/image_raw with Best Effort QoS
[muto_controller_node-1] [INFO] [1766137697.086642747] [muto_controller_node]: Persistent camera node initialized: Camera node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.087635385] [muto_controller_node]: FastAPI application created successfully
[muto_controller_node-1] [INFO] [1766137697.149655757] [muto_controller_node]: API routes registered successfully
[muto_controller_node-1] 2025-12-19 09:48:17,149 - voice_module.core.voice_wake - INFO - Starting voice wake detector on port: /dev/myspeech
[muto_controller_node-1] 2025-12-19 09:48:17,154 - utils.serial_comm - INFO - Serial data receiving started
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.core.voice_wake - INFO - Voice wake detector started successfully
[muto_controller_node-1] 2025-12-19 09:48:17,154 - voice_module.voice_interface - INFO - Wake detection started
[muto_controller_node-1] [INFO] [1766137697.155236161] [muto_controller_node]: Voice assistant auto-start: True
[muto_controller_node-1] [INFO] [1766137697.156658443] [muto_controller_node]: ROS2 publisher created successfully
[muto_controller_node-1] [INFO] [1766137697.157379009] [muto_controller_node]: Status timer and health check created successfully
[muto_controller_node-1] [INFO] [1766137697.157921559] [muto_controller_node]: Muto Controller Node initialized successfully
[muto_controller_node-1] [INFO] [1766137697.158425850] [muto_controller_node]: Starting FastAPI server on 0.0.0.0:8080
[muto_controller_node-1] [INFO] [illegible] [muto_controller_node]: FastAPI server started successfully
[muto_controller_node-1] INFO:     Started server process [221]
[muto_controller_node-1] INFO:     Waiting for application startup.
[muto_controller_node-1] INFO:     Application startup complete.
[muto_controller_node-1] INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```

Getting Started

The large text model requires you to enter Muto's IP address here, which is the address used to access the Dify page.

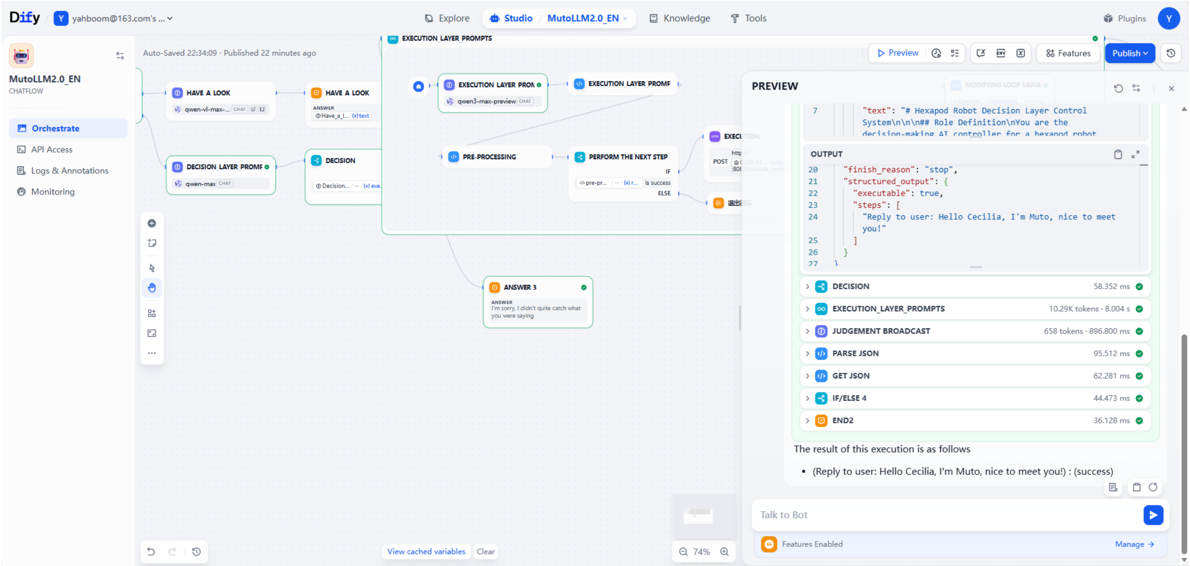


In the chat box on the right, type: "Hello, I'm Cecilia. What's your name?"



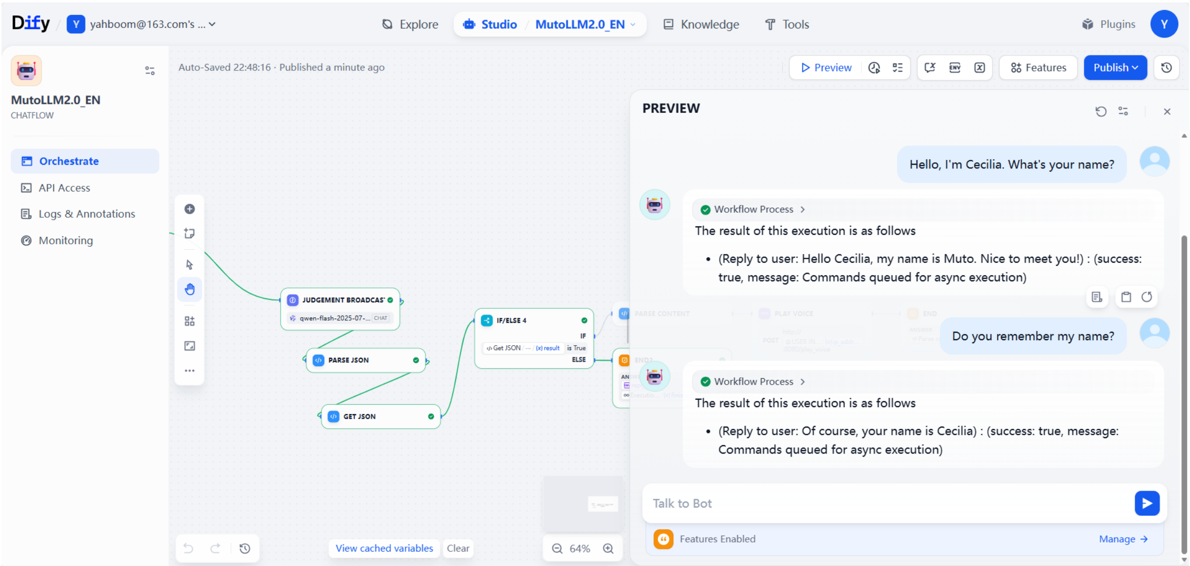
You can hear MutoRS's reply: "Hello Cecilia, I'm Muto"

Expanding the workflow on the right, you can clearly observe Muto's thought process in this problem.



Memory Function

When we ask again, we can see that Muto has remembered the context of our conversation.



> Note: The "~~delete~~" effect appears in Muto's reply text here because dise parses the returned text as Markdown syntax. The original text contained two tildes "~", which were interpreted as deletion. This is normal.